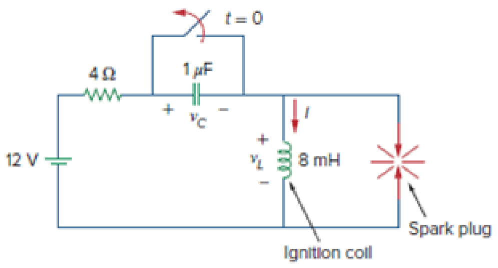
Chapter 8, Problem 16PP

Problem

In Fig. 8.52, find the capacitor voltage *vC* for *t* > 0.

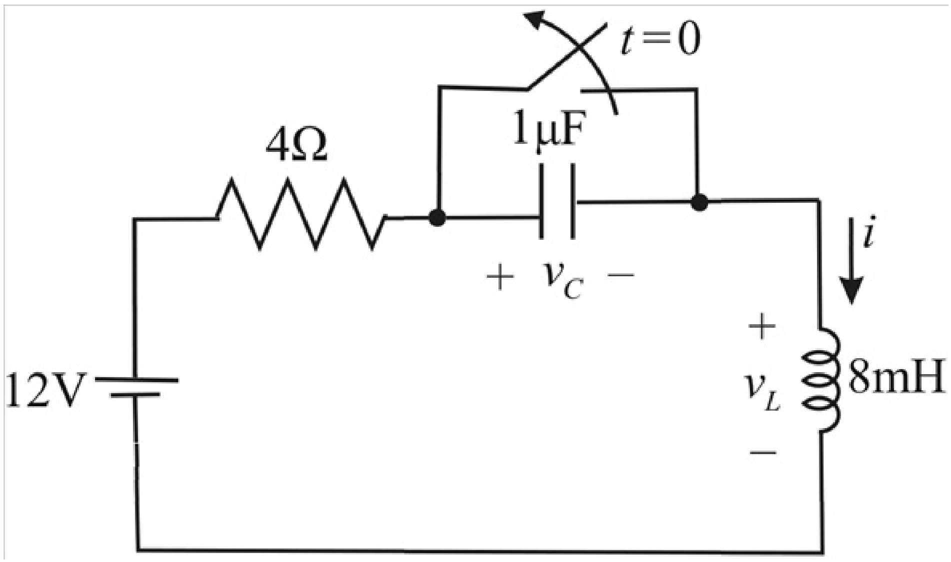
Figure 8.52:



Step-by-step solution

Step 1 of 13

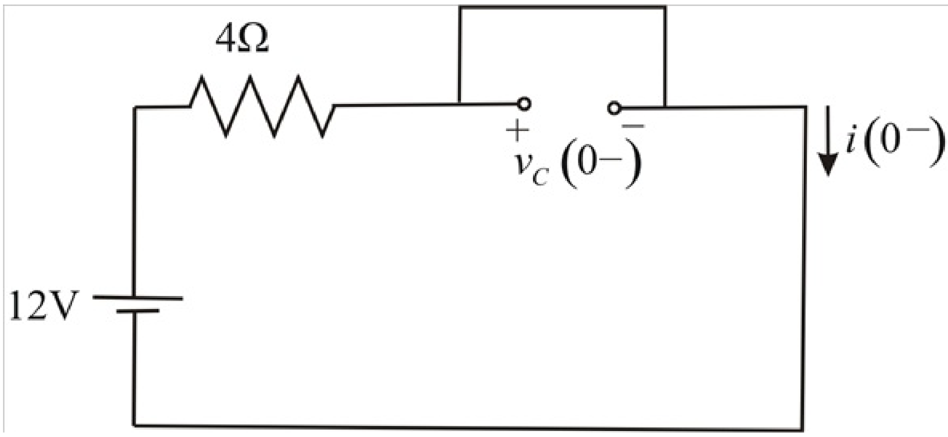
The given circuit is,



Step 2 of 13

The switch is closed for $t<0$ and the circuit is in steady state, therefore the capacitor is replaced with an open circuit and the inductor is replaced with a short circuit as shown,

Step 3 of 13



Step 4 of 13

$v_C(0^-)=0\text{V}$

$i(0^-)=\frac{12}{4}$

$i(0^-)=3\text{A}$

Due to the continuity of the capacitor voltage and inductor current,

$v_C(0^+)=v_C(0^-)$

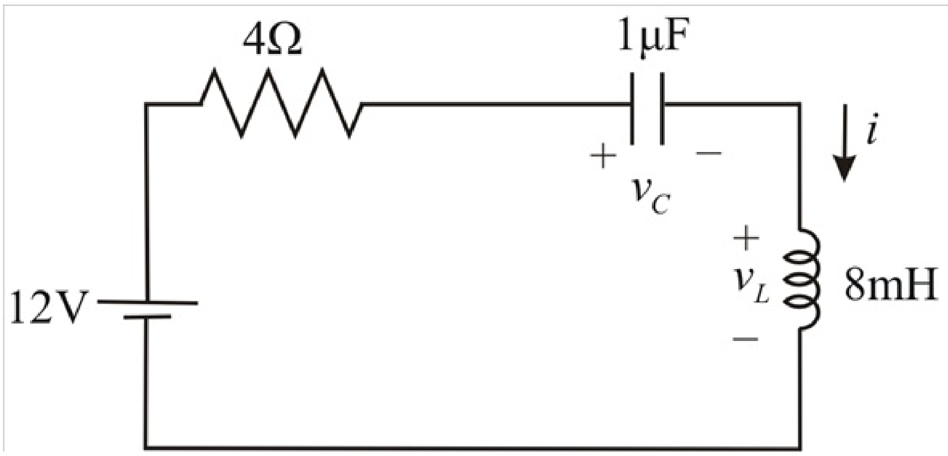
$v_C(0^+)=0\text{V}$

$i(0^+)=i(0^-)$

$i(0^+)=3\text{A}$

Step 5 of 13

For $t>0$, the switch opened, therefore the equivalent circuit is,



Step 6 of 13

The current through the capacitor at $t=0^+$ is

$$i(0^+)=C\frac{dv_C(0^+)}{dt}$$

$$\frac{dv_C(0^+)}{dt}=\frac{i(0^+)}{C}$$

$$\frac{dv_C(0^+)}{dt}=\frac{3}{1\mu}$$

$$\frac{dv_C(0^+)}{dt}=3\times10^6\ \text{V/s}$$

Step 7 of 13

The circuit is a series RLC circuit,

$$\alpha=\frac{R}{2L}$$

$$\alpha=\frac{4}{2(8\times10^{-3})}$$

$$\alpha=250$$

$$\alpha^2=62500$$

Step 8 of 13

$$\omega_0=\frac{1}{\sqrt{LC}}$$

$$\omega_0=\frac{1}{\sqrt{(8\times10^{-3})(1\times10^{-6})}}$$

$$\omega_0^2=125\times10^6$$

Since $\omega_0^2>\alpha^2$, the response is under damped therefore the form of the transient response is

$$v_{Ct}(t)=e^{-250t}(B_1\cos\omega_d t+B_2\sin\omega_d t)$$

$$\omega_d=\sqrt{\omega_0^2-\alpha^2}$$

$$\omega_d=11180$$

Therefore

$$v_{Ct}(t)=e^{-250t}(B_1\cos 11180t+B_2\sin 11180t)$$

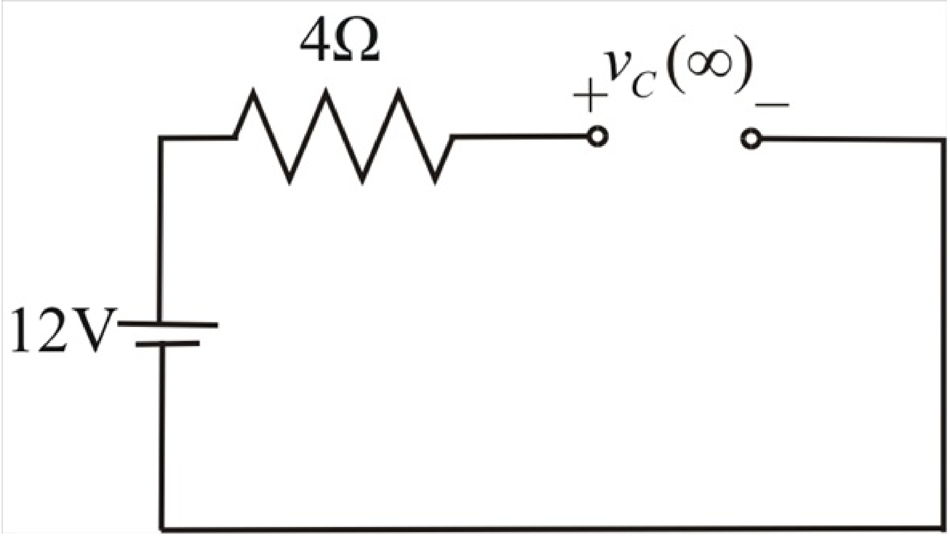
Step 9 of 13

The steady state response is,

$$v_{Css}(t)=v_C(\infty)$$

As $t\to\infty$, the circuit reaches steady state condition and the capacitor is replaced by an open circuit and the inductor is replaced by a short circuit,

Step 10 of 13



Step 11 of 13

From above circuit,

$v_C(\infty)=12\text{V}$

Hence $v_{Css}(t)=12\text{V}$

Therefore, the complete response,

$v_C(t)=v_{Ct}(t)+v_{Css}(t)$

$v_C(t)=12+e^{-250t}(B_1\cos 11180t+B_2\sin 11180t)$ .... (1)

Step 12 of 13

Using initial conditions, at $t=0$

$v_C(0)=12+B_1$

$0=12+B_1$

$B_1=-12$

Taking the derivative of $v_C(t)$ in equation (1), we get

$$\frac{dv_C(t)}{dt}=-250e^{-250t}(B_1\cos 11180t+B_2\sin 11180t)$$
$$+e^{-250t}(-11180B_1\sin 11180t+11180B_2\cos 11180t)$$

Step 13 of 13

At $t=0^+$

$$\frac{dv_C(0^+)}{dt}=-250B_1+11180B_2$$

$$3\times10^6=-250(-12)+11180B_2$$

$$11180B_2=3\times10^6-3000$$

$$B_2=268.07$$

The total response is,

$$\therefore \boxed{v_C(t)=\left(12-12e^{-250t}\cos 11180t+268.07e^{-250t}\sin 11180t\right)\ \text{V}}$$